MARKING SCHEME

# Model Paper — 06

Third Term Test (Full Syllabus)

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**Note (for teachers):**

- Part A: 2 marks for each question (generally 1 mark for the correct method and 1 mark for the correct final answer). — Total 40 marks
- Part B: 10 marks for each question. — Total 60 marks. The marks for each step are shown in the right-hand column.
- Award full marks for correct alternative methods. Accept a reasonable range for values read from graphs.

## Part A — Answers

**1.** $\sqrt{1.2321} = \sqrt{\frac{12321}{10000}} = \frac{111}{100}$

**Final answer: 1.11**

(2 marks)

**2.** $\frac{20}{8} - \frac{15}{8} = \frac{5}{8}$ ; $\frac{8}{5} \times \frac{5}{8}$

**Final answer: 1**

(2 marks)

**3.** $2a^2 - 8a + 3a - 12$

**Final answer: $2a^2 - 5a - 12$**

(2 marks)

**4.** $(2x)^2 - 2(2x)(3) + 3^2$

**Final answer: $(2x - 3)^2$**

(2 marks)

**5.** $x(x + 3)$ ; $(x + 3)(x - 3)$ ; $2x$

**Final answer: $2x(x + 3)(x - 3)$**

(2 marks)

**6.** $\frac{(x + 1)^2 - x(x - 1)}{x(x + 1)} = \frac{x^2 + 2x + 1 - x^2 + x}{x(x + 1)}$

**Final answer: $\frac{3x + 1}{x(x + 1)}$**

(2 marks)

**7.** Arc length = $\frac{1}{2} \times 2 \times \frac{22}{7} \times 7 = 22$ cm ; Perimeter = 22 + 14

**Final answer: 36 cm**

(2 marks)

**8.** $A\hat{C}B = x \Rightarrow B\hat{A}C = 2x$ ; $3x = 90° \Rightarrow x = 30°$

**Final answer: 60°**

(2 marks)

**9.** $x - 3 = 2^3 = 8$

**Final answer: x = 11**

(2 marks)

**10.** $\log_2 \frac{48}{3} = \log_2 16 = \log_2 2^4$

**Final answer: 4**

(2 marks)

**11.** $a = 4, d = 5$ ; $T_n = 4 + (n - 1)5$

**Final answer: $T_n = 5n - 1$**

(2 marks)

**12.** $A \cup B = \{a, b, c, d, e\}$

**Final answer: $(A \cup B)' = \{f\}$**

(2 marks)

**13.** $P \times (1 + \frac{18}{100}) = 14\,160 \Rightarrow 1.18P = 14\,160$

**Final answer: Rs. 12 000**

(2 marks)

**14.** Total man-days = 360 ; in 6 days = 90 ; remaining 270 ÷ 12

**Final answer: $22\frac{1}{2}$ days**

(2 marks)

**15.** Time = 60 ÷ 40 + 60 ÷ 60 = 1.5 + 1 = 2.5 hours ; 120 ÷ 2.5

**Final answer: 48 km/h**

(2 marks)

**16.** $\frac{1}{u} = \frac{1}{f} - \frac{1}{v} = \frac{v - f}{fv}$

**Final answer: $u = \frac{fv}{v - f}$**

(2 marks)

**17.** Sum 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) ⇒ 6 ; 6 ÷ 36

**Final answer: $\frac{1}{6}$**

(2 marks)

**18.** $240 \times \frac{75}{360}$

**Final answer: 50**

(2 marks)

**19.** Parallelogram = 2 × triangle = 2 × 30

**Final answer: 60 $cm^2$**

(2 marks)

**20.** 6 m = 600 cm ; 600 ÷ 200

**Final answer: 3 cm**

(2 marks)

## Part B — Answers and Mark Allocation

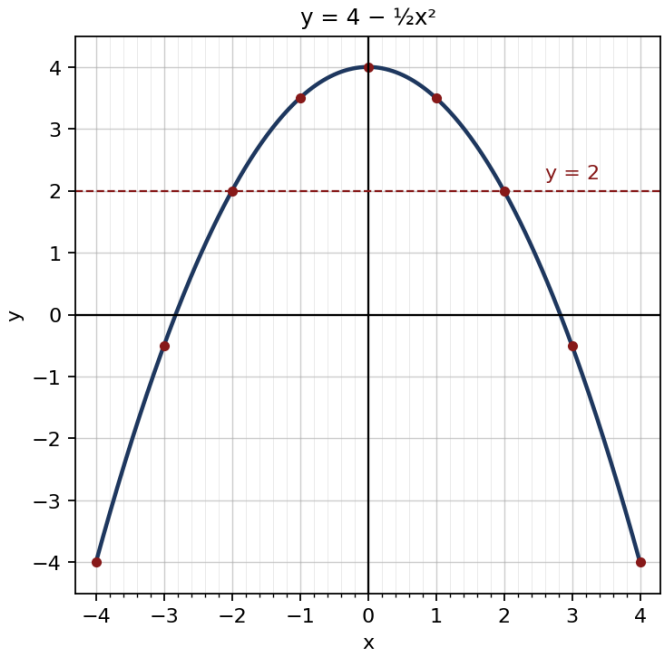
### Question 1

| | |
|---|---|
| **(a)** | |
| $(x - 5)(x + 5)$ ; $2x^2 - 9x - 5 = (2x + 1)(x - 5)$ | **1** |
| $\frac{(x - 5)(x + 5)}{(2x + 1)(x - 5)} \times \frac{2x + 1}{x + 5}$ | **1** |
| $= 1$ | **1** |
| **(b)** | |
| $f = 4s$ .....① | **1** |
| $f + 5 = 3(s + 5)$ .....② | **1** |
| $4s + 5 = 3s + 15 \Rightarrow s = 10$ | **1** |
| Son 10 years, father 40 years | **1** |
| **(c)** | |
| $(2x - 3)(x + 4) = 0$ | **2** |
| $x = \frac{3}{2}$ or $x = -4$ | **1** |

**Total marks: 10**

## Question 2

| | |
|---|---|
| **(a)** | |
| $x = -2 \Rightarrow y = 2$ ; $x = 3 \Rightarrow y = -0.5$ | **1** |
| **(b)** | |
| Correct scale and axes | **1** |
| Plotting the 9 points correctly | **1** |
| Smooth curve | **1** |




| | |
|---|---|
| **(c)** | |
| (i) Maximum value 4 ; axis of symmetry $x = 0$ | **1** |
| (ii) $x \approx \pm 2.8$ (accept $\pm 2.8$ – $\pm 2.9$) | **1** |
| $x^2 = 8 \Rightarrow x = \pm\sqrt{8}$ ; $\sqrt{8} \approx 2.8$ | **1** |
| (iii) $-2 < x < 2$ | **1** |
| (iv) $y \approx 2.9$ (accept 2.8 – 2.9) | **1** |
| **(d)** | |
| $y = -\frac{1}{2}x^2$ | **1** |

**Total marks: 10**

## Question 3

| | |
|---|---|
| **(a)** | |
| $14 \times 14 \times 10 = 1\,960\ \text{cm}^3$ | **1** |
| **(b)** | |
| $\frac{22}{7} \times 3.5 \times 3.5 \times 10$ | **1** |
| $= 385\ \text{cm}^3$ | **1** |
| **(c)** | |
| $1\,960 - 385 = 1\,575\ \text{cm}^3$ | **1** |
| **(d)** | |
| Total surface of the cuboid $= 2(14 \times 14 + 14 \times 10 + 14 \times 10) = 952\ \text{cm}^2$ | **1** |
| Removing the two circular faces $= 2 \times \frac{22}{7} \times 3.5 \times 3.5 = 77\ \text{cm}^2$ | **1** |
| Curved surface of the hole $= 2 \times \frac{22}{7} \times 3.5 \times 10 = 220\ \text{cm}^2$ | **1** |
| Total surface $= 952 - 77 + 220 = 1\,095\ \text{cm}^2$ | **1** |
| **(e)** | |
| $1\,095 \times 2$ | **1** |
| = Rs. 2 190 | **1** |

**Total marks: 10**

## Question 4

| | |
|---|---|
| **(a)** | |
| BD = DC and the height from A to BC is common to both (AD is a median) | **1** |
| $\therefore \Delta ABD = \Delta ADC$ | **1** |
| **(b)** | |
| BE is a median of ABD $\Rightarrow \Delta BED = \frac{1}{2} \Delta ABD$ | **1** |
| CE is a median of ADC $\Rightarrow \Delta CED = \frac{1}{2} \Delta ADC$ | **1** |
| $\Delta BEC = \Delta BED + \Delta CED = \frac{1}{2}(\Delta ABD + \Delta ADC) = \frac{1}{2} \Delta ABC$ | **1** |
| **(c)** | |
| $\Delta ABD = \frac{1}{2} \times 64 = 32 \text{ cm}^2$ | **1** |
| $\Delta ABE = \frac{1}{2} \times 32 = 16 \text{ cm}^2$ | **1** |
| **(d)** | |
| $\Delta FBC$ and $\Delta ABC$ stand on the same base BC and lie between the parallel lines BC $\parallel$ AF | **1** |
| $\Rightarrow \Delta FBC = 64 \text{ cm}^2$ | **1** |
| FD is a median $\Rightarrow \Delta FBD = \frac{1}{2} \times 64 = 32 \text{ cm}^2$ | **1** |

**Total marks: 10**

## Question 5

| | |
|---|---|
| **(a)** | |
| Mode = 2 books | **1** |
| **(b)** | |
| Cumulative frequencies: 3, 10, 22, ... ; the $20^{th}$ and $21^{st}$ values are 2 | **1** |
| Median = 2 books | **1** |
| **(c)** | |
| fx: 0, 7, 24, 27, 24, 15 | **1** |
| $\Sigma fx = 97$ | **1** |
| Mean = 97 ÷ 40 = 2.425 | **1** |
| **(d)** | |
| More than 3 books = 6 + 3 = 9 | **1** |
| $\frac{9}{40} \times 100 = 22.5\%$ | **1** |
| **(e)** | |
| Since 2 is added to every value, 2 is added to the mean as well | **1** |
| 2.425 + 2 = 4.425 | **1** |

**Total marks: 10**

## Question 6

| | |
|---|---|
| **(a)** | |
| $a = 25, d = -2$ ; $T_6 = 25 + 5(-2)$ | **1** |
| $= 15$ | **1** |
| **(b)** | |
| $25 + (n - 1)(-2) = 1$ | **1** |
| $n = 13$ | **1** |
| **(c)** | |
| $S_{13} = \frac{13}{2}(25 + 1)$ | **1** |
| $= 13 \times 13$ | **1** |
| $= 169$ | **1** |
| **(d)** | |
| $S_8 = \frac{8}{2}[2 \times 25 + 7 \times (-2)] = 4 \times 36 = 144$ | **1** |
| Logs removed $= 169 - 144$ | **1** |
| $= 25$ | **1** |

**Total marks: 10**